

CHG 569

Environmental Pollution Engineering

Module VII
Solid Waste Management



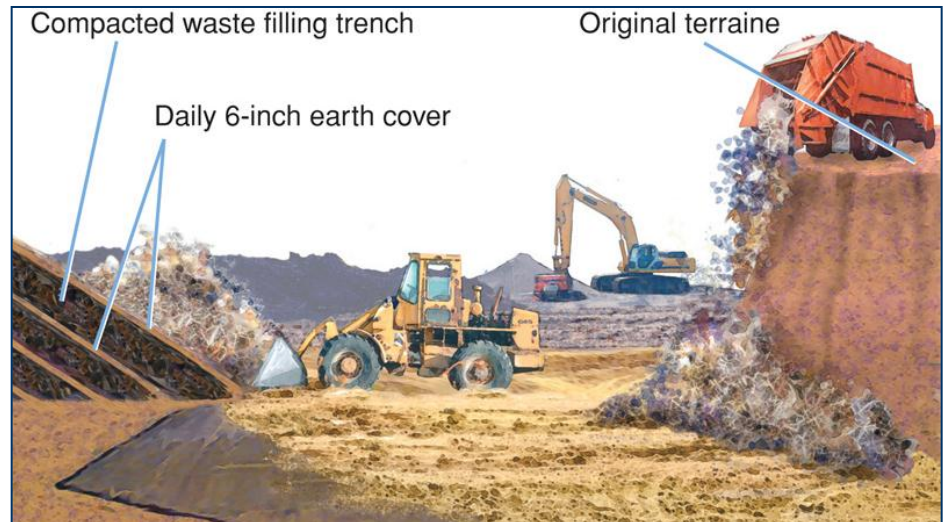
Solid Waste

These differ from air and water pollutants since these waste remain at the point of origin until a decision is made to collect and dispose of them



General approach to solid waste

- **Open dumping**
pest and health hazard
– flies, roaches, rats can transmit
all kinds of diseases
- **Sanitary landfill**
where the refuse deposited,
compacted and covered with
earth at the end of the day's
operation.
 - Area method
 - Trench method



Problems associated with landfills

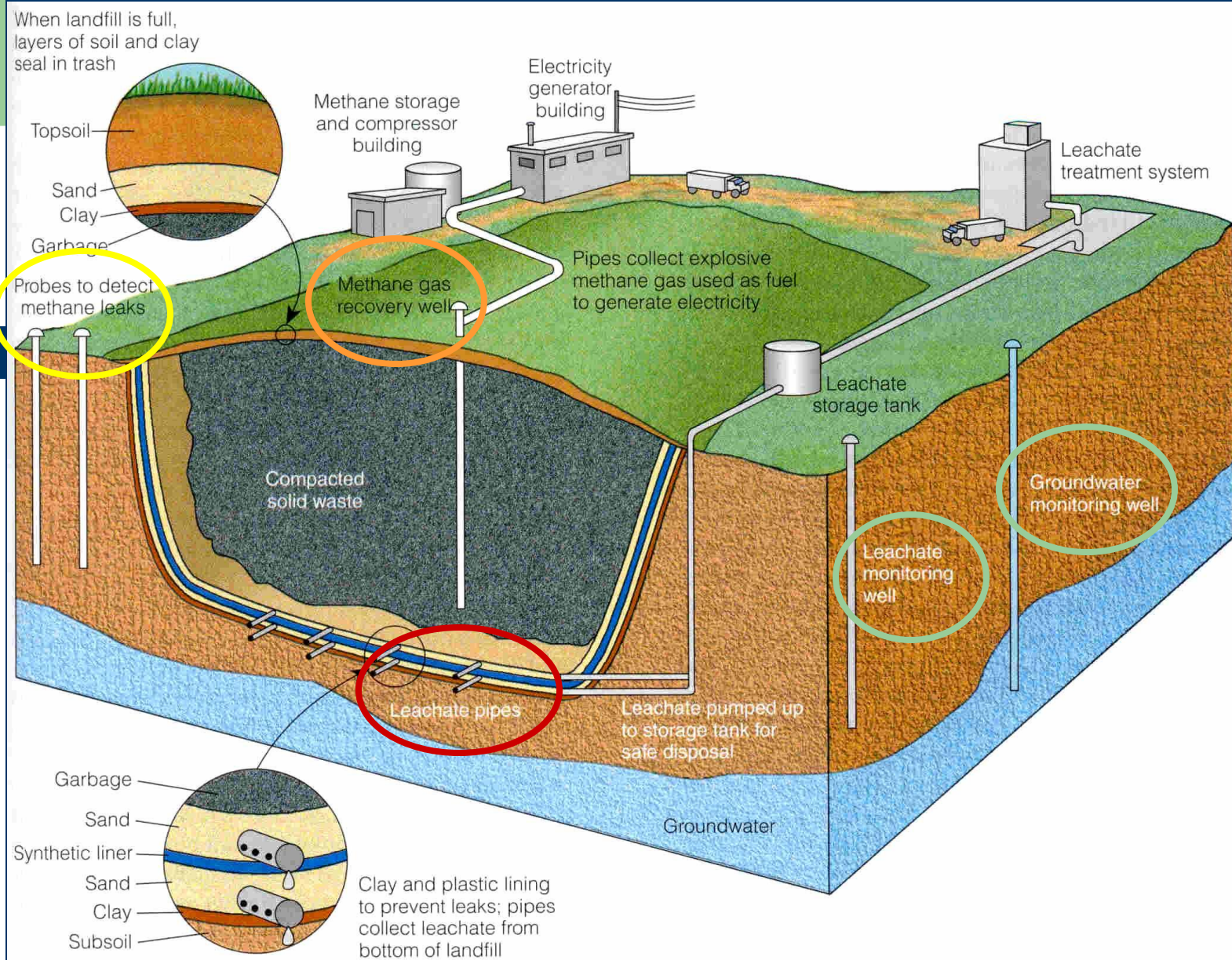
- **Leachate** is formed from rain and surface water that percolate through the mass of waste dissolving soluble solid compounds from the waste, along with some of the liquid products formed during decomposition of the waste
- **Gas** is also generated when any biodegradable material is landfilled in the presence of water

A typical landfill site

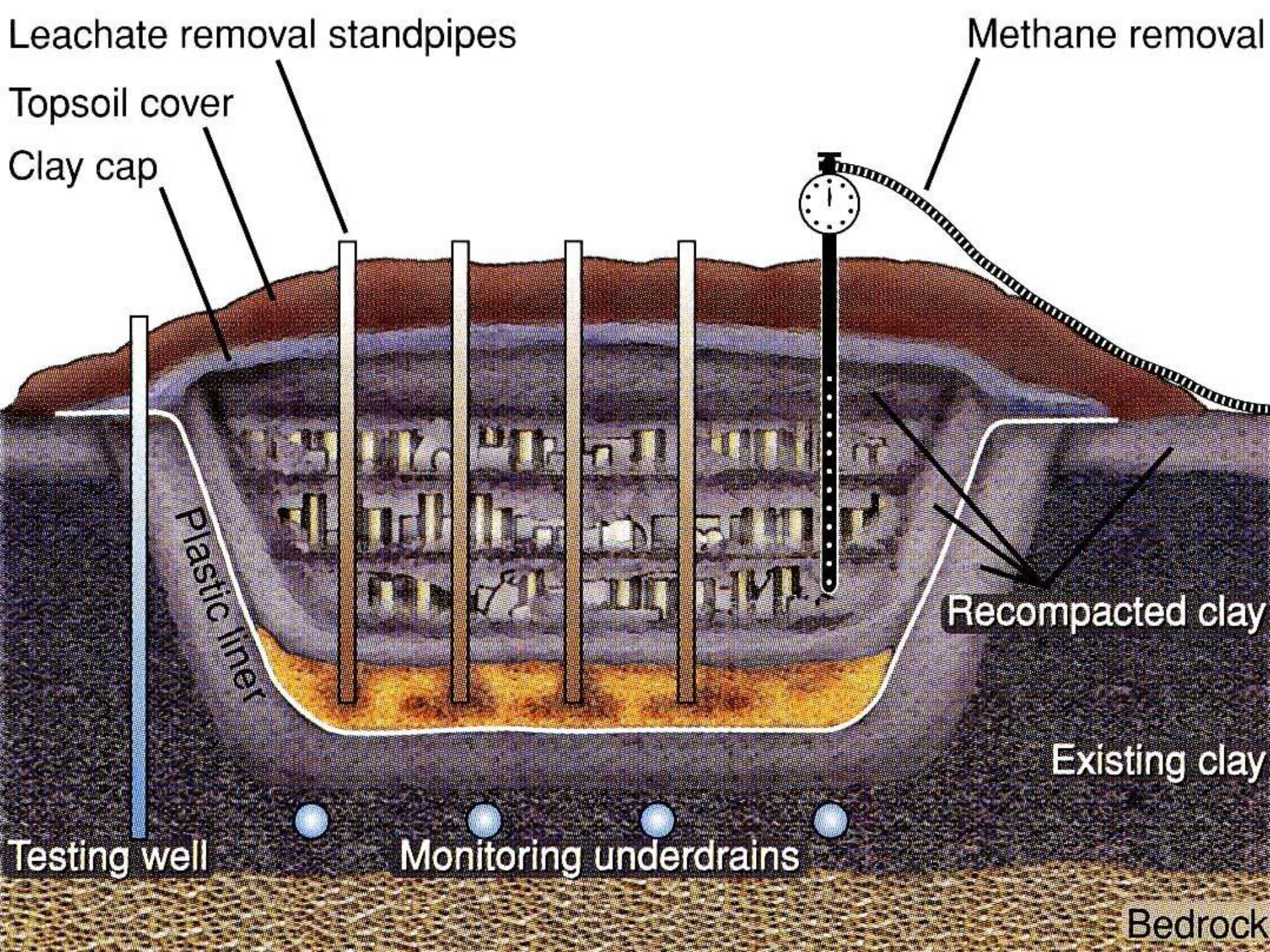


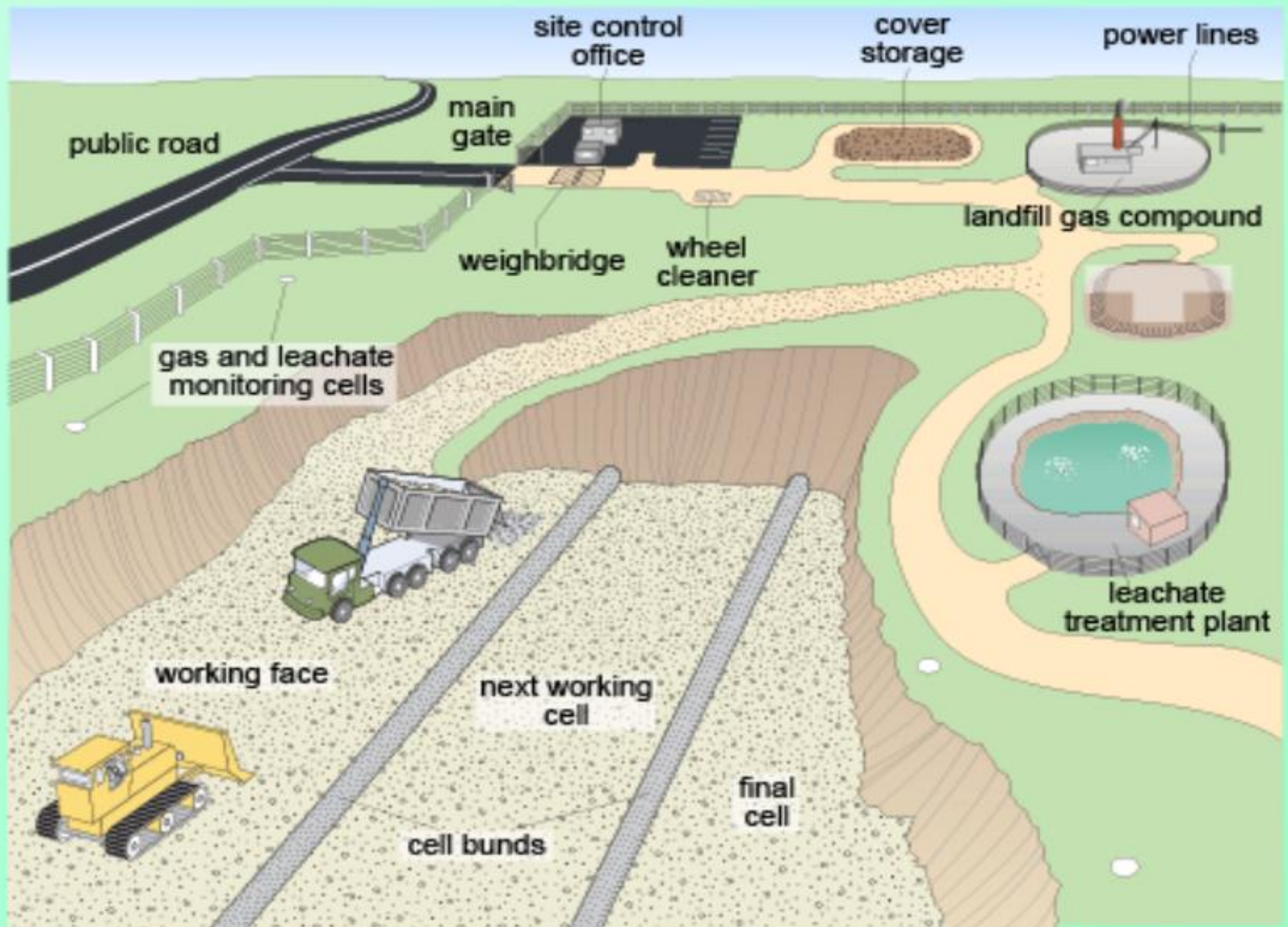
A – Completed cell, B- Prepared and lined cell, C- leachate/run-off pond

Sanitary Landfills



State-of-the-art sanitary landfills are designed to eliminate or minimize environmental problems that plague older landfills. Even such state-of-the-art landfills are expected to leak eventually, passing both the effects of contamination and cleanup costs on to future generations.





Once the landfill is full

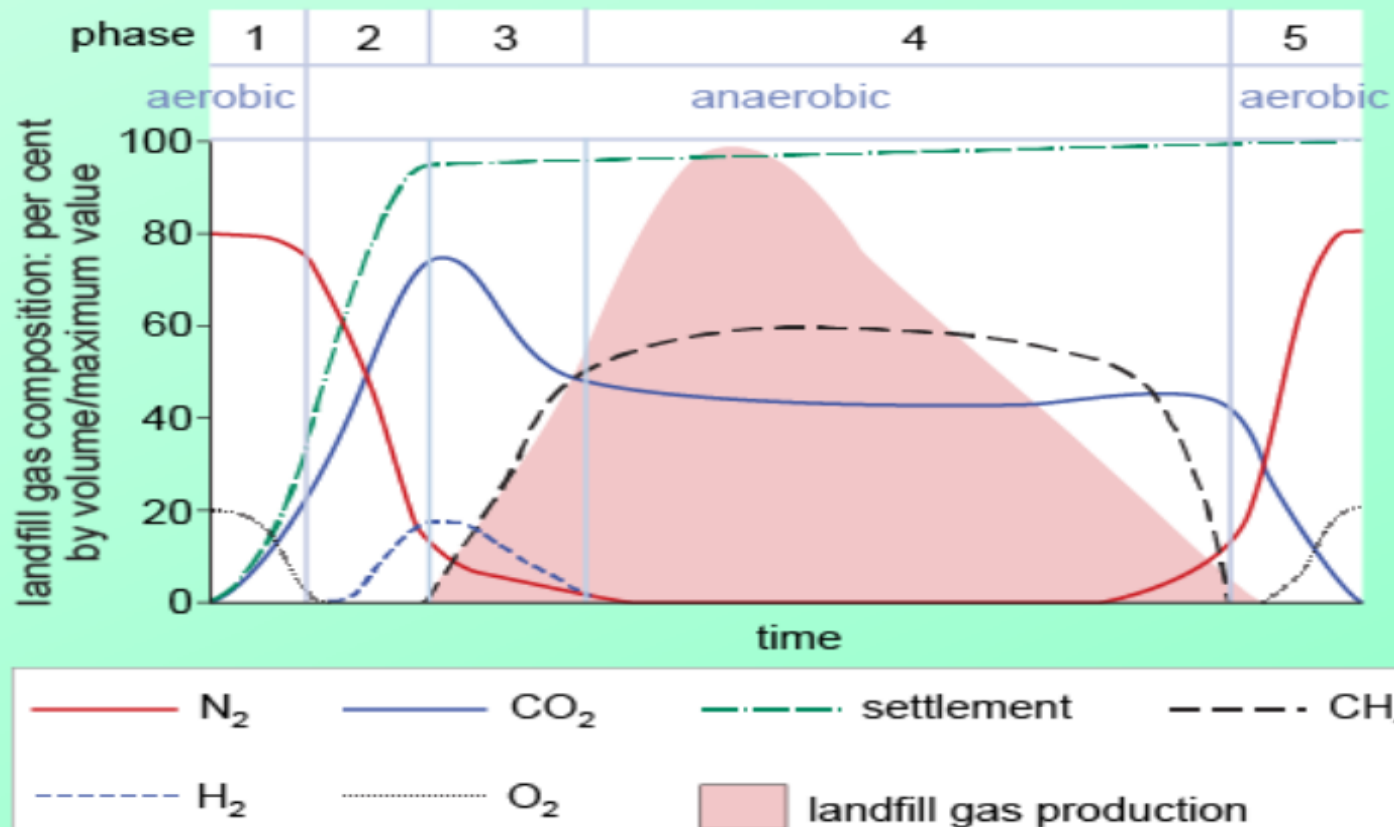
Final site cap - to minimize water ingress (and hence leachate formation) and fugitive emissions of gas, on completion of a particular landfill it is capped with an impervious layer (Clay most frequently used). Cap thickness is approx. 1m .

It takes 10 – 30 yrs after site completion for it to settle

Soil coverage - the cap is covered in soil (approx. 1m) and vegetation (grass, not vegetation with deep roots that can endanger the cap)

After use – after site has stabilised it can be used as sports ground, golf courses, open storage, or vehicle parking lot.

Landfill gas composition against time (next slide for description)



Description of slide 6

- Phase 1 –Aerobic** - oxygen level falls from 20% to zero, and CO₂ rises from 0 to around 20%, N₂ drops and settlement rises from zero to around 30%
- Phase 2 – Anaerobic** - CO₂ continues to rise to a peak of about 75%, H₂ is generated, reaches a peak of about 20%, N₂ drops to about 15% and settlement rises to 95%
- Phase 3** –CO₂ falls steeply and then gradually to about 50%, H₂ falls back to 0, N₂ falls to about 5% . Methane begins to appear and rises to around 50% at the end of this stage.
- Phase 4** –(occurs 3 - 6months after dumping waste) Methane continues to rise to about 60%, where it remains for most of this phase (15 – 30 yrs) before it drops to zero , CO₂ is fairly constant at around 40%, N₂ is about zero for most of this phase, and begins to climb as methane level falls. (Small quantities can be produced for up to 100yrs)
- Phase 5** – in the final stage, aerobic conditions become established, with O₂ levels rising from 0 to 20% and CO₂ reducing to 0. N₂ climbs back to 80% and settlement completes its gradual rise to 100%.

By the end of phase 5, the landfill is fully stabilized and its potential for pollution has ceased

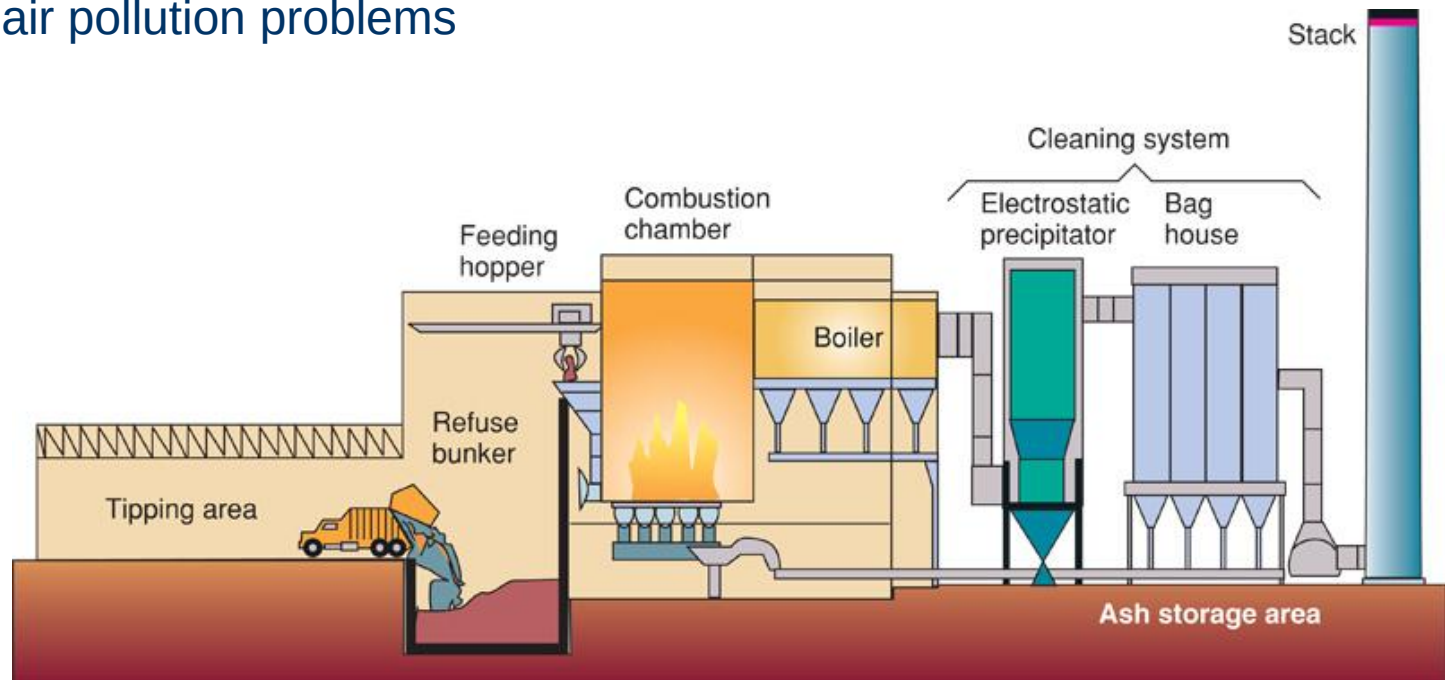
General approach to solid waste - contd

- Incineration

Reduces disposal volume by 80 -90%

Residual ash sent to landfill and usually contains toxic materials

Creates air pollution problems

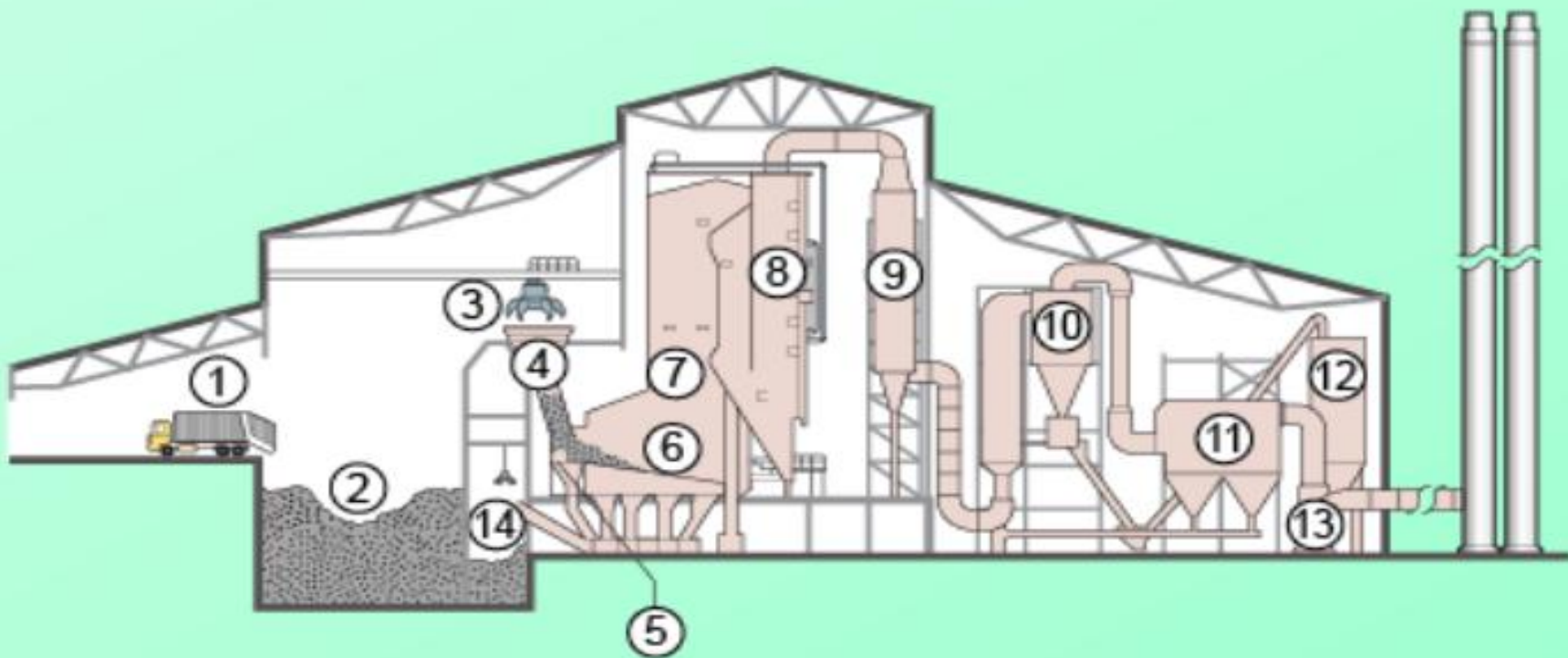


Typical values for incineration

Temperature - 1150 - 1450°C

Residence time – 2 secs

Amount of air - 1.8 – 2.5 times stoichiometric



- | | |
|--|------------------------------------|
| 1 Tipping area | 8 Superheater |
| 2 Waste storage pit | 9 Economiser |
| 3 Overhead loading crane | 10 Lime scrubbing reactor |
| 4 Feed hopper | 11 Fabric filters |
| 5 Feed grate to push waste onto combustion grate | 12 Scrubber residue storage hopper |
| 6 Combustion grate | 13 Fan |
| 7 Secondary combustion zone/boiler | 14 Grate ash storage |

Gas cleaning from incineration –

to prevent turning a solid waste problem to an air pollution problem

Air from the waste boiler

- SNCR (where ammonia or urea is added to the gas stream –this converts NO_x to Nitrogen)
- HEX (for cooling),
- lime and Activated carbon are added (lime reacts with SO₂ to form gypsum and the AC traps all kinds of formed species, esp dioxins and furans)
- Clean gas to the atmosphere via a stack.

Anaerobic Digestion of organic waste

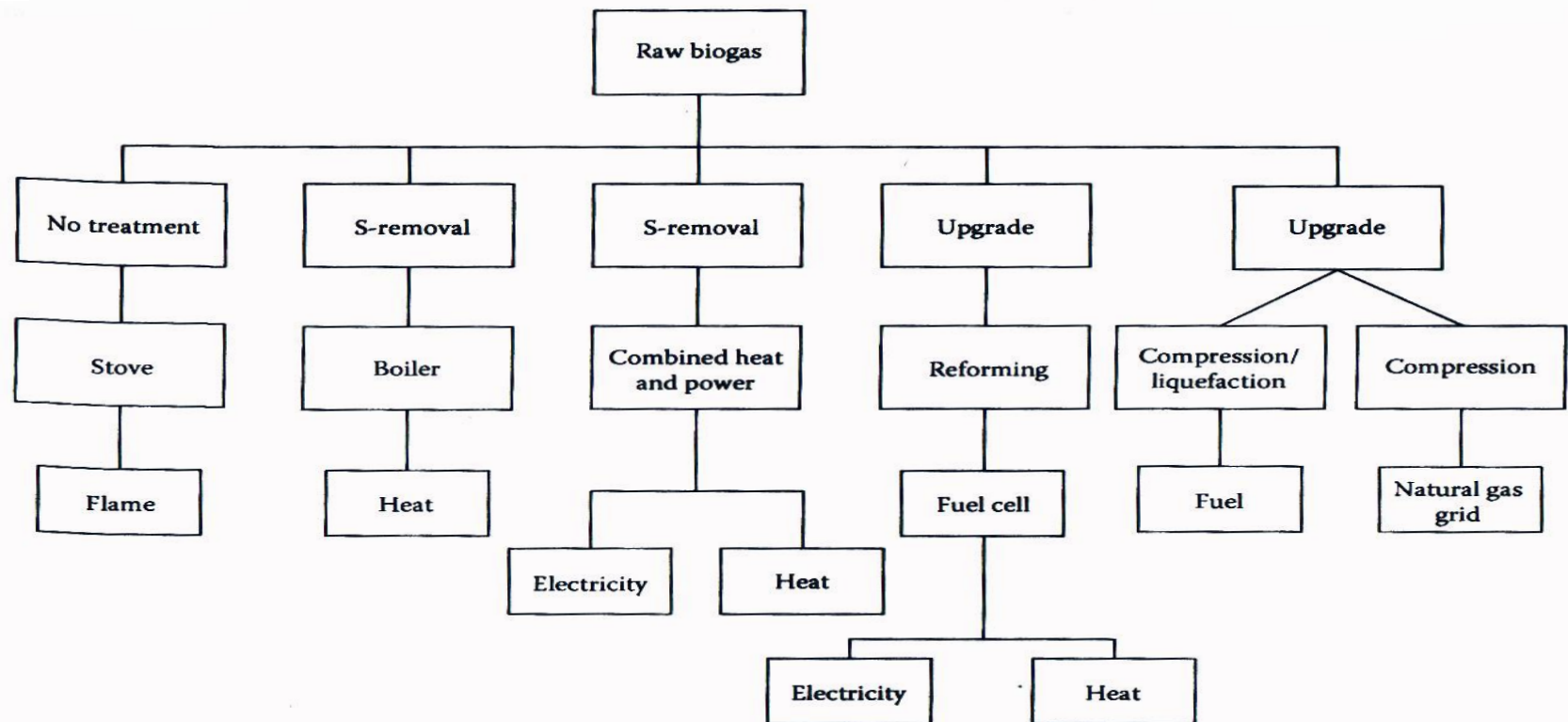
AD of organic waste gives 55 -77% methane, along with CO_2 , H_2S , NH_3 , particles and water. Depending on the end-use of the biogas (see slide 8), post treatment (see slide 9) might be necessary to upgrade the biogas to a purity of >97%.

CO_2 is of no energy value

H_2S is very corrosive

Humidity can easily be removed via a water trap (a U-bend tube filled with water)

Biogas utilization purposes and required upgrading methods



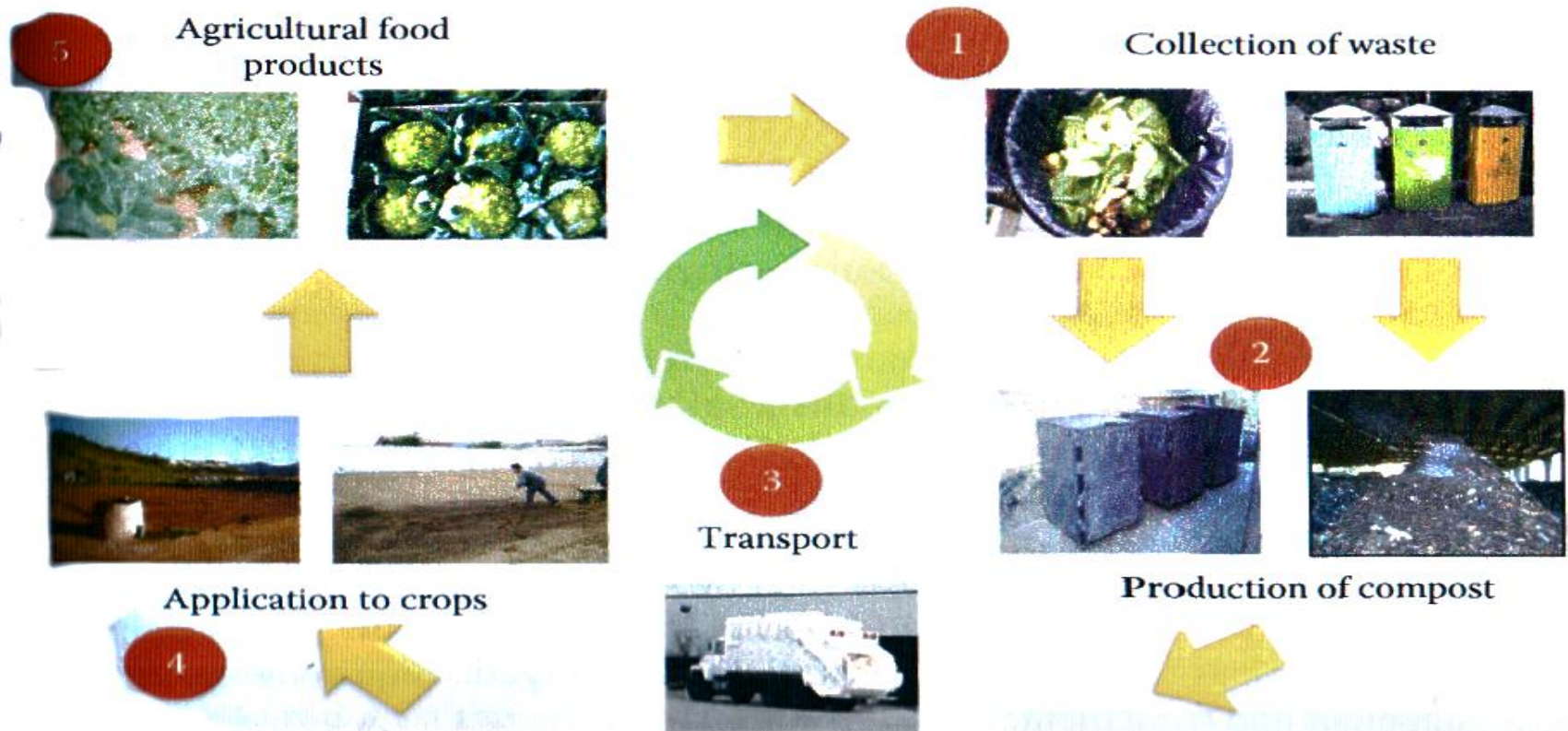
Common biogas upgrading Technologies

Separation Method	Process	Principle	Final CH ₄ Content
Physical absorption	Pressurized water wash	Dissolution of CO ₂ in water at high pressure	>96%
	Wash in solvent	Dissolution of CO ₂ in solvent	>96%
Adsorption	Pressure swing adsorption	Adsorption of CO ₂ on a molecular sieve	>96%
Chemical absorption	Monoethanolamine (MEA) wash	Chemical reaction between CO ₂ and MEA	>99%
Cryogenic process	Low temperature process	CO ₂ goes to liquid phase, while CH ₄ remains gaseous	>99.9%
Membrane separation	Polymer membrane gas separation (dry)		>80%
	Membrane gas separation (wet)	Membrane permeability of CH ₄ is lower than H ₂ S and CO ₂	>96%

Composting- used for food waste

2V [Vegetable-to-Vegetable Model]- composting closes the organic matter cycle

Organic vegetable recycle helps achieve zero waste



Recycle/Reuse

Metals are an essential part of modern society – buildings, vehicles, household utilities, equipment, industrial plants, machines, electronic devices, etc, etc

Metals come from mineral ore which are natural resources mined from the earth's crust, and to get 1 unit of metal, 100 units of ore might need to be processed.

Metals tend to have an “eternal” life cycle, unlike other products that depreciate with each recycle/reuse.

General approach to solid waste - contd

- Recycle/Reuse

Energy requirements: Virgin vs Recycle

Metal	Virgin Material Requirement (kwh/ton)	Recycled Material Requirement (kwh/ton)	kwh saving per ton of recycled material	Barrels of oil Saved per ton of recycled material
Aluminium	51379	2000	49379	29.1
Copper	13532	1727	11805	7.0
Iron	4270	1666	2604	1.5
Magnesium	90821	1875	88946	52.2
Paper	6730	2520	4210	2.5
Titanium	126115	52416	73699	43.3

Recovery of materials from solid waste pays, as recycling saves so much energy e.g. recycled aluminum (soft drink cans) uses only 2 -3 % of the energy required to make new aluminum from bauxite

Waste reduction/reuse pays

The University buys 4200 reams (11 tonnes) of paper each year at a cost of N84,000 per ream. Of this, 30% ends up in letters, reports, etc. that are delivered to staff or stored. The rest is disposed of to a LAWMA that charges N4500 per tonne to collect and landfill the paper. Compare the cost of the following scenarios:

1. the current situation
2. the introduction of double-sided printing for reports where possible, resulting in paper savings of 10%
3. The situation in part (2), plus sending the scrap paper to a recycling company that pays N1000 per tonne for this type of paper

Solution

1. The annual cost of buying the paper is:

$$N84000 \times 11 = N924000$$

the annual cost of disposing of 70% of it is:

$$N4500 \times 11 \times 0.7 = N34650, \text{ say } N34700$$

So the total cost of scenario (1) is $N924000 + N34700 = \text{N958700 per year}$

2. The University's paper consumption falls to 90% of its previous level, so its annual purchasing cost reduces to:

$$N924000 \times 0.9 = N831600$$

there is also an equivalent saving in the annual disposal cost

$$N34700 \times 0.9 = N31230, \text{ say } N31200$$

so the total cost of scenario (2) is $N831600 + N31200 = \text{N862800 per year}$. Which is an annual **savings of N95900** compared with scenario (1)

3. The cost of buying the paper remains N831600 per year, but instead of a disposal cost of N31200 there is now an annual income of:

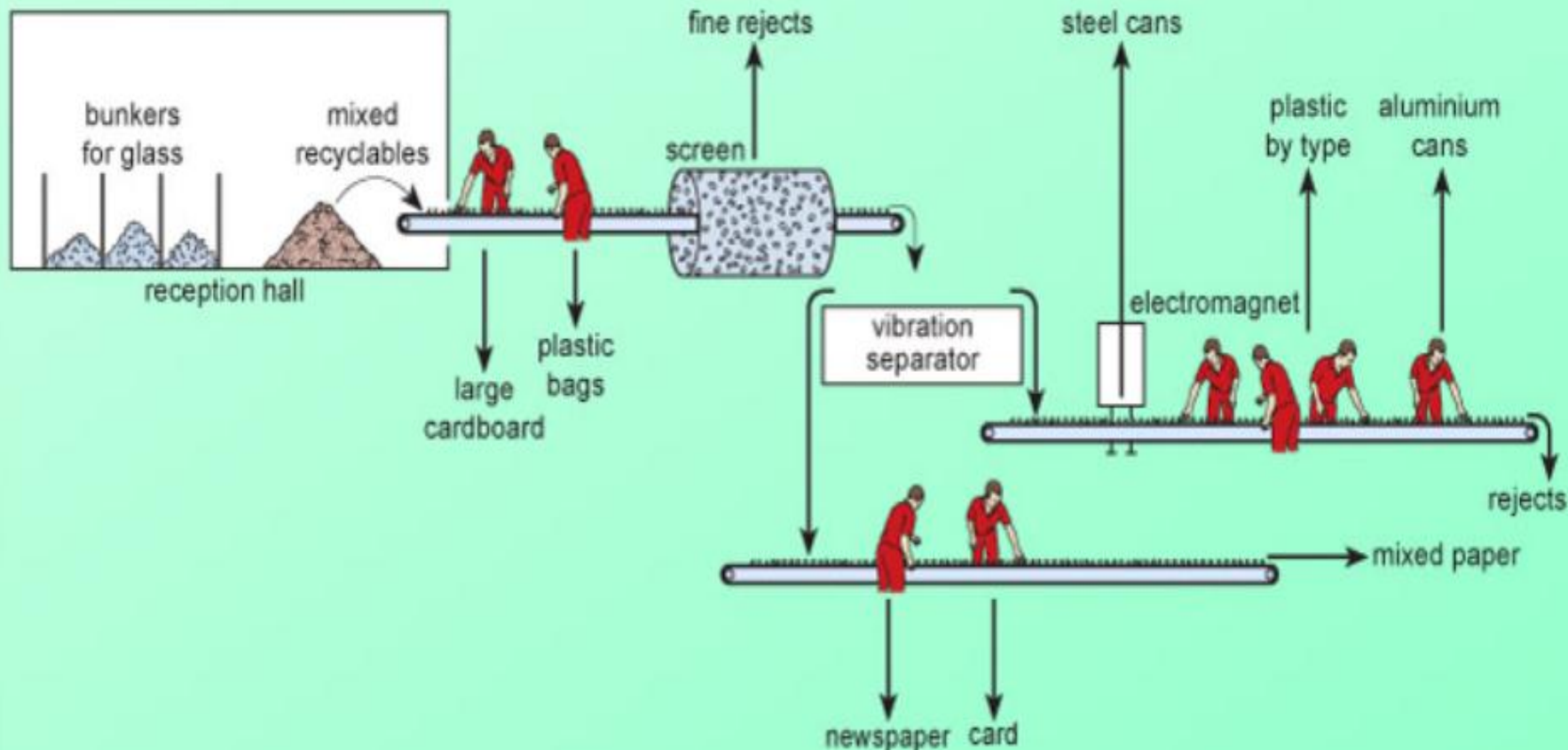
$$N1000 \times 0.9 \times 11 \times 0.7 = N6930, \text{ say } N6900$$

so the total cost of scenario (3) is $N831600 - N6900 = \text{N824700 per year}$. This represents an annual **saving of N134000** compared with scenario (1), or approximately 14%

No longer waste management but resource management,
that is, think not of it as something to be disposed of but
as a resource to be used

Waste management → Resource Management

Taking products one sector no longer needs and
making them into useful resources for another sector



Waste-2-wealth videos

https://drive.google.com/drive/folders/1sthdVgvtCS1o43FJrBsgKBzf8BTe3rPz?usp=share_link

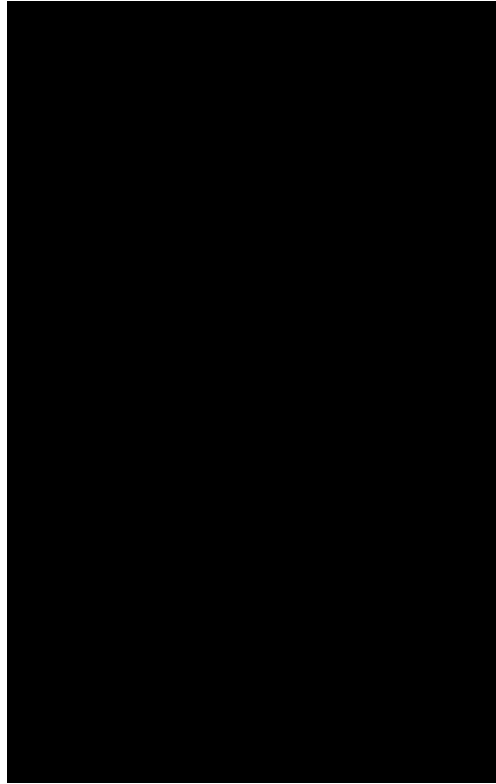
-  banana stem waste to wealth.mp4
-  Banana trash to cash.MP4
-  Chopsticks to furniture_1_the beginning.mp4
-  Chopsticks to furniture_2_more products.mp4
-  Module 7_xtra_behind...-waste-into-fabric.pdf
-  Plastic waste to plasti...ks to build schools.mp4
-  sculpting waste tyres.mp4
-  waste materials to useful products.pdf
-  Waste materials to useful products.pptx
-  waste nylons to tiles.mp4
-  waste to wealth sche...nana stem to paper.mp4

A clip on waste to wealth



Go online for more waste to wealth ideas

Using banana stem as raw material - 2



Using waste nylon as raw material



Using plastic waste as raw material



Other waste to wealth schemes –

it's all about 'waste' recovery and utilization . Who is best to do it, if not the chemical engineer, who adds value to raw materials to get useful products.

Invasive aquatic weeds such as water hyacinth, cat-tail, water- lily, water lettuce can be converted to useful products

Mitimeth (visit handle on instagram and facebook) uses such for handcraft items
(leaves and roots are used for paper making)

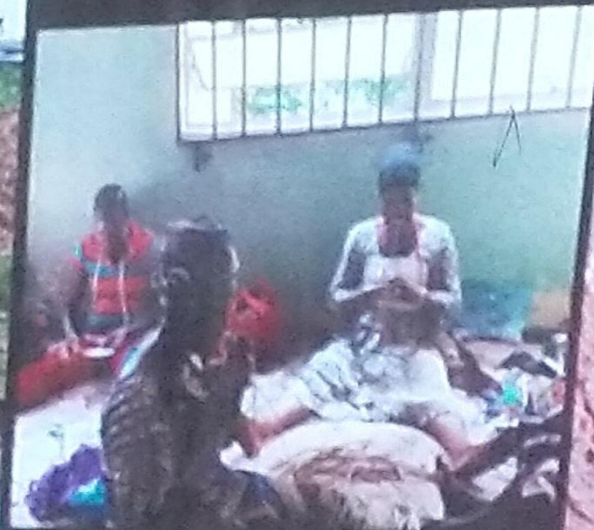
Visit you-tube videos – Handcrafts from water hyacinth in Vietnam

Next slide for more pics.

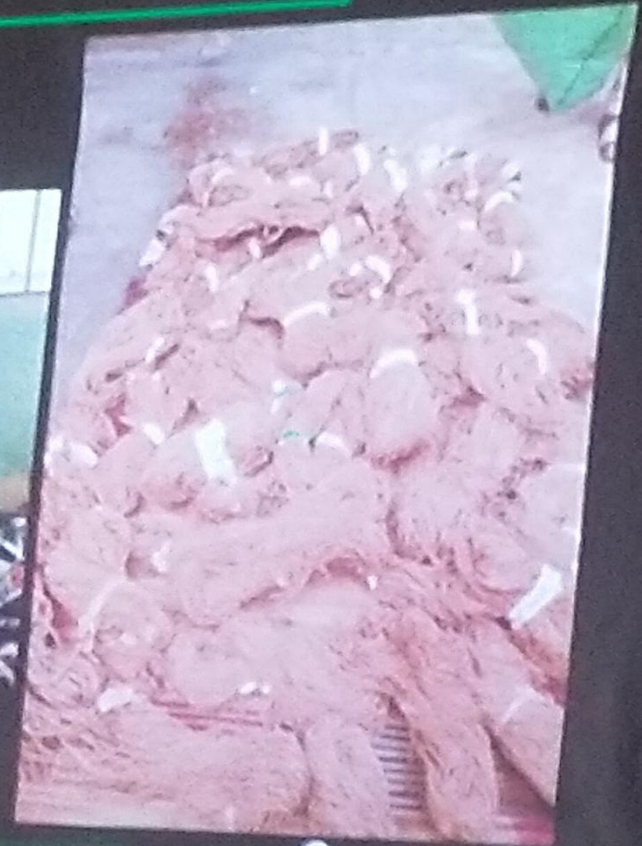


Water Hyacinth – an invasive aquatic weed

The Transformation Process



2



3

Harvesting, drying, weaving

Drying ->dried weed - > woven weed



Products



‘The World in your hand ’sculpture



Sculpting waste tyres

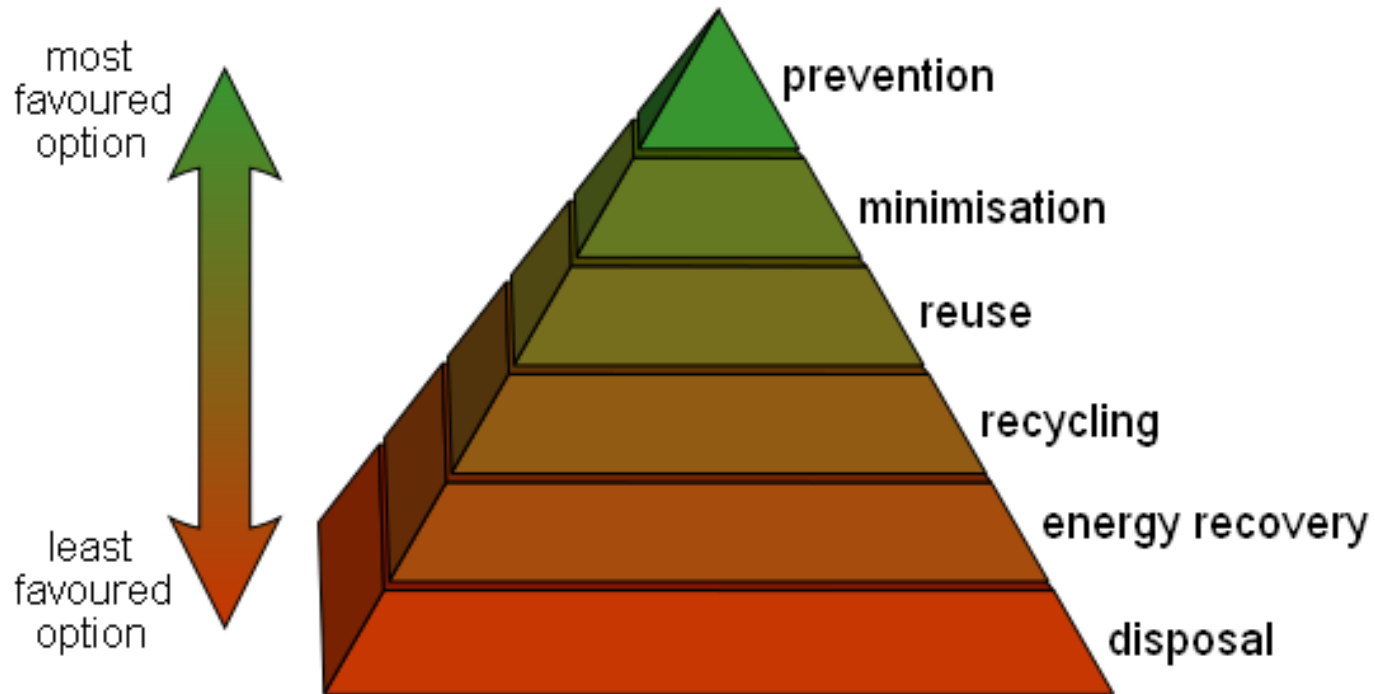


Waste tyres as a resource

In the past twenty years, many attempts have been made to utilise recycled waste tyre rubber as an aggregate substitute in concrete and pavements



Hierarchy of management options



Industrial waste Management

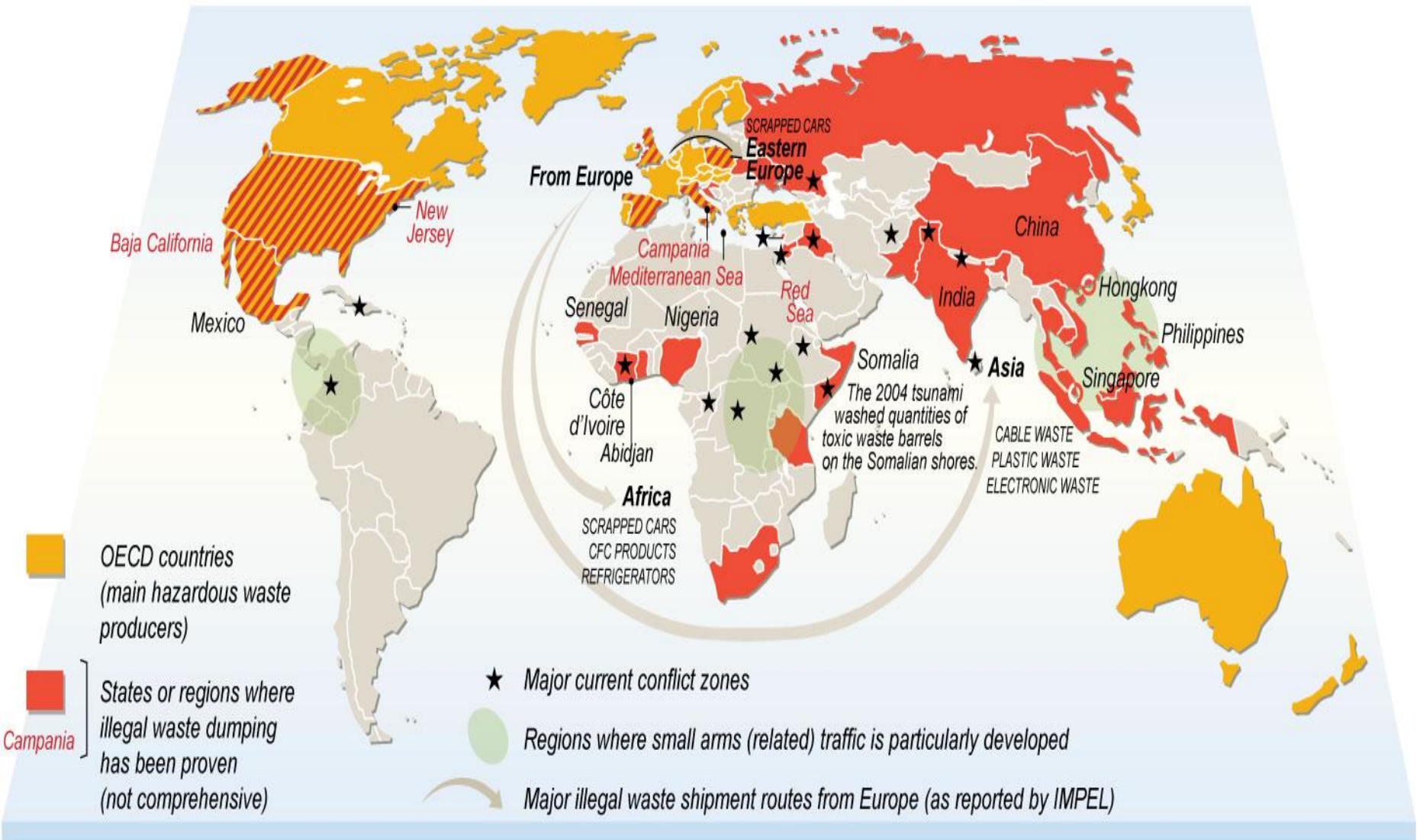
Industrial waste tend to be hazardous because they are either:

- ignitable (i.e., flammable)
- reactive
- Corrosive, or
- Toxic – either mutagenic, carcinogenic, or teratogenic to humans or other life forms

This type of waste is either:

- diluted and dispersed, or
- concentrated and confined (NIMBY) – dumped out of sight – ocean dumping or shipped to less developed countries- waste trafficking (see next slide)

Trafficking waste stories



OECD – Organisation for Economic cooperation and development

IMPEL – EU network for the implementation and enforcement of Environmental Law

HAZARDOUS WASTE MANAGEMENT OPTIONS

- Produce Less Waste
 - Avoid creating wastes in the first place
 - Reuse and Recycle
- Convert to Less Hazardous Substances
 - Physical Treatment (Tie-up or Isolation)
 - Incineration
 - Chemical Processing (Transformation)
 - Bioremediation (Microorganisms)
- Store Permanently
 - Retrievable Storage - Can be inspected and periodically retrieved.
 - Secure Landfills - Modern, complex landfills with multiple liners and other impervious layers and monitoring systems.

SOLVING THE ENVIRONMENTAL PROBLEM

HOW?

Zero Waste means no pollution

Waste is unwanted product
and that is why it is

disposed off into the environment

There has to be a **paradigm shift on waste**, it must be seen as a **resource
that can be put to use**

(more of this in Sustainable Development)